

Proficiency test 2

Vincent Zheng

Task 1

```
for (i in 1:3){  
  print(mtcars$mpg[i])  
}
```

```
[1] 21  
[1] 21  
[1] 22.8
```

Task 2

```
count_above <- function(vector, value){ #creates the function that takes two inputs  
  count <- 0 #creates an object that acts as a counter  
  for (i in 1:length(vector)){ #for loop that runs however long the vector is  
    if (vector[i] > value){ #runs if the vector value is greater than the cutoff  
      count <- count + 1 #increases the count object by 1  
    }  
  }  
  return(count) #returns the count value  
}
```

```
count_above(iris$Sepal.Length, 6)
```

```
[1] 61
```

```
count_above(mtcars$disp, 150)
```

```
[1] 20
```

Task 3

Masking functions is when a package that was newly installed contains a function that shares a name with a function that has either existed in a previously installed package or the base r functions. As a result, if not specifically specified, when running the function name, the one that corresponds to the newest package will be ran. For example, if package 'w' has a function called 'g' and has been installed and package 'q' was installed after that also has a function named 'g', the function g from package q will be ran when it is called unless w was libraried in the code chunk.

Task 4

```
iris$Sepal.Length[1] #returns the sepal length of the first iris flower in the dataset
```

```
[1] 5.1
```

```
iris[1,1]
```

```
[1] 5.1
```